Algebraic structures of the Lindblad equation

Example 5

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Abstract

This Mathematica notebook accompanies the paper Algebraic Structures of the Lindblad Equation and presents five worked examples illustrating the algebraic formulation of finite-dimensional Lindblad dynamics.

Three examples verify the derived algebraic relations for one-, two-, and three-qubit operator bases. The calculations proceed from the construction of an orthonormal Hermitian operator basis and its structure constants to the associated superoperator algebra. The resulting Liouville superoperators and differential equations for the dynamical-map coefficients are compared with those obtained by direct construction, confirming their agreement.

Two numerical examples describe transmons truncated to their two and four lowest energy levels, including coherent driving, energy relaxation, and pure dephasing. They solve the dynamical-map equations and illustrate the resulting population dynamics. The four-level example demonstrates an approximate X gate and population leakage beyond the computational subspace. Both examples construct the unnormalized Choi matrix to check Hermiticity preservation, complete positivity, and trace preservation of the computed maps. Additional checks in the four-level example examine purely dissipative evolution with the Hamiltonian set to zero and unitary evolution with the Kossakowski matrix set to zero.

The algebraic construction is parameterized by the number of qubits and includes recursion relations for extending it to larger operator bases. Although symbolic calculations and storage requirements become demanding as the system size increases, the implementation provides a framework that can be adapted to other finite-dimensional open quantum systems.

Licence

LindbladAlgebra Mathematica examples accompanying the paper

“The Algebraic Structure of the Lindblad Equation”

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In this section we create a series of Mathematica functions that facilitate the computation of the matrix basis, structure constants and Liouville superoperator.

Instructions

Run the cell Functions below before running any example. Running Example 2 requires running first Example 1 in order to have the needed mathematical expressions. Running Example 4 requires running first Example 3

Functions

In this section we create a series of Mathematica functions that facilitate the computation of the matrix basis, structure constants and Liouville superoperator.

Functions

Example 5. Three qubits.

Three-qubit matrix basis h

In the following example we verify the main relations derived in the paper for the two qubit case.

In[•]:=
```
nq = 3;
```

We begin by constructing the one-qubit basis $\mathbf{h}_n$ and its associated structure constant superoperators $\boldsymbol{Z}$, $\boldsymbol{Z}^*$ and $\boldsymbol{X}$.

In[•]:=
```
h = QubitBasis[nq];
{b, c} = BCStructureConstantsFromQubits[nq];
z = ZStructureConstantsFromQubits[nq];
zt = Conjugate[z];
x = XBasisFromQubits[nq];
```

The complex structure-constant superoperators can be visualized with the following line of code.

In[•]:=
```
MatrixForm[z[[3]]]
```

Out[•]//MatrixForm=

$$\begin{pmatrix}
0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & -\frac{i}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
\frac{1}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{i}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{i}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{\sqrt{2}} & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{\sqrt{2}} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{\sqrt{2}} \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{\sqrt{2}} & 0 & 0
\end{pmatrix}$$

$$\begin{pmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{\sqrt{2}} & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
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0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{pmatrix}$$

```
0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0  0
```

It can be verified that $\boldsymbol{Z}$ and $\boldsymbol{Z}^*$ constitute to independent realizations of the Lie algebra associated with h by showing that the following difference, obtained by subtracting the right-hand side from the left-hand side of Eqs. (70) and (71), vanish identically.

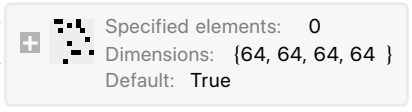
In[•]:=
```
zz = Transpose[z.Transpose[z], {1, 3, 2, 4}];
commu = SparseArray[zz - Transpose[zz]];
PossibleZeroQ[SparseArray[commu + 2 ⅈ c.z]]
```

Out[•]=

SparseArray[]

In[•]:=
```
ztzt = Transpose[zt.Transpose[zt], {1, 3, 2, 4}];
commu = SparseArray[ztzt - Transpose[ztzt]];
PossibleZeroQ[SparseArray[commu - 2 ⅈ c.zt]]
```

Out[•]=

SparseArray[]

Similarly we can verify that $\boldsymbol{Z}$, $\boldsymbol{Z}^*$ commute.

```
In[•]:= zzt = Transpose[z.ConjugateTranspose[z]];
        ztz = Transpose[zt.Transpose[z], {3, 1, 2, 4}];
        commu = SparseArray[zzt - ztz];
        PossibleZeroQ[commu]
```

```
Out[•]=
```

SparseArray[ Specified elements : 0
Dimensions: {64, 64, 64, 64 }
Default: True]

The structure of the elements of $\boldsymbol{X}$ can be visualized using the following line of code.

```
In[•]:= MatrixForm[x[[2, 3]]]
```

```
Out[•]//MatrixForm=
```

$$\begin{pmatrix}
0 & 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
-\frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{i}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{8} & 0 & 0 & 0 & 0 & 0 & 0
\end{pmatrix}$$

$$\begin{pmatrix}
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -\frac{i}{8} & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0

$$\begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

For some computations it is convenient to flatten the first two indices of X.

In[•]:=
```
X = Flatten[x, 1]
```

Out[•]=

SparseArray [Specified elements : 262 144 Dimensions: {4096, 64, 64 }]

Data not saved. Save now

Hamiltonian

A convenient system for illustrating the results developed in this work is the transmon qubit. Since the number of quantum levels retained in the model can be chosen arbitrarily, it provides a simple framework for studying the behavior of the algebraic structures introduced above as the Hilbert-space dimension increases. We first define the lowering and rasing operators as well as the jump operators

In[•]:=
```
m = 2^nq;
n = 2^2 nq;
a = SparseArray[Table[{n + 1, n + 2} → Sqrt[n + 1], {n, 0, m - 2}], m];
at = SparseArray[Table[{n + 1, n} → Sqrt[n], {n, 1, m - 1}], m];
A = SparseArray[Table[{n + 1, n + 1, n + 2} → Sqrt[n + 1], {n, 0, m - 2}], {m - 1, m, m}];
At = SparseArray[Table[{n + 1, n + 2, n + 1} → Sqrt[n + 1], {n, 0, m - 2}], {m - 1, m, m}];
```

The Hamiltonian in the interaction picture, given by Eq. (129) is calculated using the following commands

In[•]:=
```
ψ = Exp[-i (Δ t + ϕ)] Map[Exp, i α t (1 - Range[m - 1])];
ψt = Exp[i (Δ t + ϕ)] Map[Exp, -i α t (1 - Range[m - 1])];
Hi = Ω/2 (ψ.A + ψt.At);
MatrixForm[Hi]
```

Out[•]//MatrixForm=

$$\begin{pmatrix}
0 & \frac{1}{2} e^{-i (t\Delta+\phi)} \Omega & 0 & 0 & 0 \\
\frac{1}{2} e^{i (t\Delta+\phi)} \Omega & 0 & \frac{e^{-i t\alpha - i (t\Delta+\phi)} \Omega}{\sqrt{2}} & 0 & 0 \\
0 & \frac{e^{i t\alpha + i (t\Delta+\phi)} \Omega}{\sqrt{2}} & 0 & \frac{1}{2}\sqrt{3}\, e^{-2 i t\alpha - i (t\Delta+\phi)} \Omega & 0 \\
0 & 0 & \frac{1}{2}\sqrt{3}\, e^{2 i t\alpha + i (t\Delta+\phi)} \Omega & 0 & e^{-3 i t\alpha - i (t\Delta+\phi)} \Omega \\
0 & 0 & 0 & e^{3 i t\alpha + i (t\Delta+\phi)} \Omega & 0 \\
0 & 0 & 0 & 0 & \frac{1}{2}\sqrt{5}\, e^{4 i t\alpha + i (t\Delta+\phi)} \Omega \\
0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0
\end{pmatrix}$$

The Hamiltonian vector is obtained by projecting the Hamiltonian on the elements of the set $\mathbf{h}_n$ through Eq. (23).

```
In[ ]:= H = ExpToTrig[ProjectOperatorOntoBasis[Hi, h]];
        MatrixForm[H]
```

Out[ ]//MatrixForm=

$$
\begin{pmatrix}
0 \\
\frac{\Omega \text{Cos}[t\,\Delta+\phi]}{2\sqrt{2}} + \frac{1}{2}\sqrt{\frac{3}{2}}\,\Omega \text{Cos}[2\,t\,\alpha + t\,\Delta + \phi] + \frac{1}{2}\sqrt{\frac{5}{2}}\,\Omega \text{Cos}[4\,t\,\alpha + t\,\Delta + \phi] + \frac{1}{2}\sqrt{\frac{7}{2}}\,\Omega \text{Cos}[6\,t\,\alpha + t\,\Delta + \phi] \\
\frac{\Omega \text{Sin}[t\,\Delta+\phi]}{2\sqrt{2}} + \frac{1}{2}\sqrt{\frac{3}{2}}\,\Omega \text{Sin}[2\,t\,\alpha + t\,\Delta + \phi] + \frac{1}{2}\sqrt{\frac{5}{2}}\,\Omega \text{Sin}[4\,t\,\alpha + t\,\Delta + \phi] + \frac{1}{2}\sqrt{\frac{7}{2}}\,\Omega \text{Sin}[6\,t\,\alpha + t\,\Delta + \phi] \\
0 \\
0 \\
\frac{1}{2}\,\Omega \text{Cos}[t\,\alpha + t\,\Delta + \phi] + \frac{1}{2}\sqrt{3}\,\Omega \text{Cos}[5\,t\,\alpha + t\,\Delta + \phi] \\
-\frac{1}{2}\,\Omega \text{Sin}[t\,\alpha + t\,\Delta + \phi] - \frac{1}{2}\sqrt{3}\,\Omega \text{Sin}[5\,t\,\alpha + t\,\Delta + \phi] \\
0 \\
0 \\
\frac{1}{2}\,\Omega \text{Sin}[t\,\alpha + t\,\Delta + \phi] + \frac{1}{2}\sqrt{3}\,\Omega \text{Sin}[5\,t\,\alpha + t\,\Delta + \phi] \\
\frac{1}{2}\,\Omega \text{Cos}[t\,\alpha + t\,\Delta + \phi] + \frac{1}{2}\sqrt{3}\,\Omega \text{Cos}[5\,t\,\alpha + t\,\Delta + \phi] \\
0 \\
0 \\
\frac{\Omega \text{Cos}[t\,\Delta+\phi]}{2\sqrt{2}} - \frac{1}{2}\sqrt{\frac{3}{2}}\,\Omega \text{Cos}[2\,t\,\alpha + t\,\Delta + \phi] + \frac{1}{2}\sqrt{\frac{5}{2}}\,\Omega \text{Cos}[4\,t\,\alpha + t\,\Delta + \phi] - \frac{1}{2}\sqrt{\frac{7}{2}}\,\Omega \text{Cos}[6\,t\,\alpha + t\,\Delta + \phi] \\
\frac{\Omega \text{Sin}[t\,\Delta+\phi]}{2\sqrt{2}} - \frac{1}{2}\sqrt{\frac{3}{2}}\,\Omega \text{Sin}[2\,t\,\alpha + t\,\Delta + \phi] + \frac{1}{2}\sqrt{\frac{5}{2}}\,\Omega \text{Sin}[4\,t\,\alpha + t\,\Delta + \phi] - \frac{1}{2}\sqrt{\frac{7}{2}}\,\Omega \text{Sin}[6\,t\,\alpha + t\,\Delta + \phi] \\
0 \\
0 \\
0 \\
0 \\
0 \\
0 \\
\frac{\Omega \text{Cos}[3\,t\,\alpha + t\,\Delta + \phi]}{\sqrt{2}} \\
-\frac{\Omega \text{Sin}[3\,t\,\alpha + t\,\Delta + \phi]}{\sqrt{2}} \\
0 \\
0 \\
-\frac{\Omega \text{Sin}[3\,t\,\alpha + t\,\Delta + \phi]}{\sqrt{2}} \\
-\frac{\Omega \text{Cos}[3\,t\,\alpha + t\,\Delta + \phi]}{\sqrt{2}} \\
0 \\
0 \\
0 \\
0 \\
0 \\
0 \\
0 \\
0 \\
0 \\
\frac{\Omega \text{Sin}[3\,t\,\alpha + t\,\Delta + \phi]}{}
\end{pmatrix}
$$

$$\begin{pmatrix}
\frac{\sqrt{2}}{\frac{\Omega \text{Cos}[3\,t\,\alpha+t\,\Delta+\phi]}{\sqrt{2}}} \\
0 \\
0 \\
\frac{\Omega \text{Cos}[3\,t\,\alpha+t\,\Delta+\phi]}{\sqrt{2}} \\
-\frac{\Omega \text{Sin}[3\,t\,\alpha+t\,\Delta+\phi]}{\sqrt{2}} \\
0 \\
0 \\
0 \\
0 \\
0 \\
0 \\
\frac{\Omega \text{Cos}[t\,\Delta+\phi]}{2\sqrt{2}} + \frac{1}{2}\sqrt{\frac{3}{2}}\,\Omega \text{Cos}[2\,t\,\alpha+t\,\Delta+\phi] - \frac{1}{2}\sqrt{\frac{5}{2}}\,\Omega \text{Cos}[4\,t\,\alpha+t\,\Delta+\phi] - \frac{1}{2}\sqrt{\frac{7}{2}}\,\Omega \text{Cos}[6\,t\,\alpha+t\,\Delta+\phi] \\
\frac{\Omega \text{Sin}[t\,\Delta+\phi]}{2\sqrt{2}} + \frac{1}{2}\sqrt{\frac{3}{2}}\,\Omega \text{Sin}[2\,t\,\alpha+t\,\Delta+\phi] - \frac{1}{2}\sqrt{\frac{5}{2}}\,\Omega \text{Sin}[4\,t\,\alpha+t\,\Delta+\phi] - \frac{1}{2}\sqrt{\frac{7}{2}}\,\Omega \text{Sin}[6\,t\,\alpha+t\,\Delta+\phi] \\
0 \\
0 \\
\frac{1}{2}\Omega \text{Cos}[t\,\alpha+t\,\Delta+\phi] - \frac{1}{2}\sqrt{3}\,\Omega \text{Cos}[5\,t\,\alpha+t\,\Delta+\phi] \\
-\frac{1}{2}\Omega \text{Sin}[t\,\alpha+t\,\Delta+\phi] + \frac{1}{2}\sqrt{3}\,\Omega \text{Sin}[5\,t\,\alpha+t\,\Delta+\phi] \\
0 \\
0 \\
\frac{1}{2}\Omega \text{Sin}[t\,\alpha+t\,\Delta+\phi] - \frac{1}{2}\sqrt{3}\,\Omega \text{Sin}[5\,t\,\alpha+t\,\Delta+\phi] \\
\frac{1}{2}\Omega \text{Cos}[t\,\alpha+t\,\Delta+\phi] - \frac{1}{2}\sqrt{3}\,\Omega \text{Cos}[5\,t\,\alpha+t\,\Delta+\phi] \\
0 \\
0 \\
\frac{\Omega \text{Cos}[t\,\Delta+\phi]}{2\sqrt{2}} - \frac{1}{2}\sqrt{\frac{3}{2}}\,\Omega \text{Cos}[2\,t\,\alpha+t\,\Delta+\phi] - \frac{1}{2}\sqrt{\frac{5}{2}}\,\Omega \text{Cos}[4\,t\,\alpha+t\,\Delta+\phi] + \frac{1}{2}\sqrt{\frac{7}{2}}\,\Omega \text{Cos}[6\,t\,\alpha+t\,\Delta+\phi] \\
\frac{\Omega \text{Sin}[t\,\Delta+\phi]}{2\sqrt{2}} - \frac{1}{2}\sqrt{\frac{3}{2}}\,\Omega \text{Sin}[2\,t\,\alpha+t\,\Delta+\phi] - \frac{1}{2}\sqrt{\frac{5}{2}}\,\Omega \text{Sin}[4\,t\,\alpha+t\,\Delta+\phi] + \frac{1}{2}\sqrt{\frac{7}{2}}\,\Omega \text{Sin}[6\,t\,\alpha+t\,\Delta+\phi] \\
0
\end{pmatrix}$$

Lindbladian

To construct the Lindbladian, we first introduce the corresponding jump operators. In the case of transmons, two decoherence processes are particularly relevant: energy relaxation and pure dephasing. Energy relaxation describes the loss of an excitation from the transmon to its environment and is represented by the Lindblad jump operator $J_1 = a$. This process is characterized by the relaxation time T_1, time or equivalently by the relaxation rate $\Gamma_1 = 1/T_1$. Although the corresponding contribution to the Lindbladian is time independent in the Schrödinger picture, it becomes time dependent in the interaction picture used here to express the Hamiltonian. For a harmonic oscillator a simple scalar phase cancels completely, but for the anharmonic transmon the phase contains the number operator n=$a^\dagger a$, that introduces the time dependency. This time dependency is easily introduced through the phases and decomposing into transition operators the lowering and rising operators

```
In[•]:= J1 = SparseArray[Table[{n, n, n + 1} → √n Exp[-i α n t], {n, 1, m - 1}], {m - 1, m, m}];
        J1t = SparseArray[Table[{n, n + 1, n} → √n Exp[i α n t], {n, 1, m - 1}], {m - 1, m, m}];
```

The Lindbladian corresponding to the energy relaxation part is defined by Eq. (38)

```
In[•]:= γ1 = SparseArray[Flatten[Table[{ii, jj} → γ10, {ii, m - 1}, {jj, m - 1}]], {m - 1, m - 1}];
        Γ1 = GammaFromgamma[h, γ1, J1, J1t];
```

Pure dephasing describes a phenomenon that does not change the energy-level populations but instead, destroys coherence between different energy eigenstates. It is characterized by the jump operator $J_2 = a^\dagger a$

```
In[•]:= J2 = SparseArray[{at.a}];
        J2t = SparseArray[{at.a}];
```

and the pure-dephasing time T_2 or rate $\Gamma_2 = 4\,T_1 - 2/T_2$ which is associated

```
In[•]:= γ2 = SparseArray[{{γ20}}];
        Γ2 = GammaFromgamma[h, γ2, J2, J2t];
        MatrixForm[Γ2]
```

```
Out[•]//MatrixForm=
```

$$\begin{pmatrix}
49\,\gamma 20 & 0 & 0 & -7\,\gamma 20 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & -14\,\gamma 20 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
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-7\,\gamma 20 & 0 & 0 & \gamma 20 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 2\,\gamma 20 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\
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\end{pmatrix}$$

$$\begin{pmatrix}
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0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0
\end{pmatrix}$$

The full Lindbladian is therefore

In[ ]:=
```
Γ = Γ1 + Γ2;
```

Algebraic Method

Having obtained the quantities required to construct the Hamiltonian and Lindbladian contributions to the Liouville superoperator, namely ***H*** and Γ, we can proceed to compute Λ using Eq. (60), yielding

In[ ]:=
```
Λ = LambdaFromHamiltonianLindbladian[z, H, Γ]
```

Out[ ]=

SparseArray [Specified elements: 736
Dimensions: {64, 64}]

Data not saved. Save now

The Liouville superoperator is obtained using Eq. (59).

In[•]:=
```
Lam = SparseArray[LiouvilleSOFromLamda[x, Λ]]
```

Out[•]=

SparseArray [Specified elements: 2816
Dimensions: {64, 64}]

Data not saved. Save now

Direct method

The Liouville superoperator is also computed in the same matrix basis for comparison. The two methods are found to yield identical results.

In[•]:=
```
Ldm = SparseArray[LiouvilleSOFromProjection[h, H, Γ]];
```

Comparison of the algebraic method and the direct method

To compare the Liouville superoperators obtained using the algebraic and direct methods, we subtract one superoperator from the other.

In[•]:=
```
PossibleZeroQ[SparseArray[Lam - Ldm]]
```

Out[•]=

SparseArray [Specified elements: 448
Dimensions: {64, 64}
Default: True]

Comparison of the algebraic method with the standard Kronecker product method

To compare the Liouville superoperators obtained using the algebraic and standard Kronecker product methods, we first construct the Kronecker representation using Eqs. (21) and (22).

In[•]:=
```
ℐ = SparseArray[ IdentityMatrix[m]];
J1T = SparseArray[Table[{n, n + 1, n} → √n Exp[-ⅈ α n t], {n, 1, m - 1}], {m - 1, m, m}];
J1s = SparseArray[Table[{n, n, n + 1} → √n Exp[ⅈ α n t], {n, 1, m - 1}], {m - 1, m, m}];
J2T = Transpose[J2, {1, 3, 2}];
J2s = Conjugate[J2];
Lsme = LiouvilleStandardKronecker[Hi, ℐ, γ1, J1, J1t, J1s, J1T, γ2, J2, J2t, J2s, J2T]
```

Out[•]=

SparseArray [Specified elements: 336
Dimensions: {64, 64}]

We then transform it from the matrix basis $\mathbf{e}_n$, defined in Eq. (12), to the basis $\mathbf{h}_n$ using the unitary transformation $u_{i,j}$=tr[$h_i e_j$]

In[•]:=
```
e = ElementaryBasis[m];
u = TensorContract[h.Transpose[e], {{2, 4}}];
ut = ConjugateTranspose[u];
Lsmh = u.(Lsme.ut)
```

Out[•]=

SparseArray [Specified elements: 2688
Dimensions: {64, 64}]

Data not saved. Save now

To compare the Liouville superoperators obtained using the algebraic and standard Kronecker product methods, we subtract one superoperator from the other.

In[•]:=
```
PossibleZeroQ[SparseArray[Lam - Lsmh]]
```

Out[•]=

SparseArray [Specified elements: 3040
Dimensions: {64, 64}
Default: True]

Liouville equation for the dynamical map

To verify that the dynamical differential equations for the coordinate vector, Eq. (113), are equivalent to those derived from the Lindbladian, we directly project the Liouville operator acting on the dynamical map, as given on the right-hand side of Eq. (110), onto the basis $\boldsymbol{X}$. We then subtract the resulting expression from the right-hand side of Eq. (113), which should yield zero if both formulations are equivalent. First, we define the elements of the dynamical map as in Eq. (108).

In[•]:=
```
V = SparseArray[Array[v[##][t] &, {n, n}]];
```

We calculate the right-hand side of the dynamical differential equations using the algebraically derived Eq. (113).

In[•]:=
```
DiffEqsAlgebraicMethod =
 SparseArray[Flatten[V].Flatten[Transpose[SparseArray[1/4 zt.Λ.z], {1, 3, 2, 4}], 1]]
```

Out[•]=

SparseArray [Specified elements: 4096
Dimensions: {64, 64}]

Data not saved. Save now

Here, we project the Liouville operator obtained through the direct method, acting on the dynamical map as given on the right-hand side of Eq. (110), onto the basis $\boldsymbol{X}$.

In[•]:=
```
DiffEqsDirectMethod = TensorContract[
  Transpose[V].TensorContract[x.Transpose[x, {2, 3, 1, 4}].Ldm, {{3, 6}}], {{1, 2}}]
```

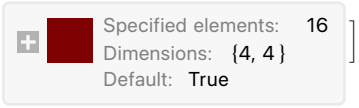
Out[•]=

SparseArray [Specified elements: 16
Dimensions: {4, 4}]

Finally, we compare the results obtained by both methods.

In[ ]:=
```
PossibleZeroQ[SparseArray[DiffEqsDirectMethod - DiffEqsAlgebraicMethod]]
```

Out[ ]=

SparseArray []